Modelling Neutron Transport and Tritium Production in a Simulated Fusion Breeder Blanket

MPhys Research Project

Rhodri Peters (10848618)

In collaboration with Vadan Khan

The University of Manchester

27th November 2025

Abstract

The continuous production of tritium is a pressing issue in deuterium-tritium (D-T) fusion research. Breeder blankets offer a potential solution. Consisting of lithium-based compounds, breeder blankets produce tritium upon interacting with fusion neutrons. Building them into the reactor structure allows for in-situ tritium breeding while also providing shielding and heat extraction. Maintaining tritium self-sufficiency is a key goal for practical breeder blankets. The goal of this project is to use a Monte Carlo simulation of neutron transport to evaluate and compare the tritium production capabilities of several breeder blanket designs. In this project we carried out systematic parameter investigations to characterise the tritium breeding behaviour of a variety of blanket structures and compositions. By simulating a homogenous blanket structure we determined that a water-cooled, liquid $Li_{17}Pb_{83}$ breeder had the highest tritium breeding performance of the materials under investigation.

1 Introduction

1.1 Nuclear Fusion

1.1.1 The D-T Fusion Reaction

Deuterium-tritium fusion has been of considerable interest since it was first proposed in 1942 by Emil Konopinski as a candidate for a fusion-based hydrogen bomb (Paris and Chadwick, 2023).

$$\mathrm{d} + \mathrm{t} \longrightarrow \mathrm{n} + \alpha + 17.6\,\mathrm{MeV} \cdot \tag{1}$$

The postwar boom in nuclear physics research led to early designs for a fusion based energy producing reactor, hoping to replicate the success of the already ubiquitous fission power plants (Azizov, 2012). The D-T reaction quickly emerged as the frontrunner for use in a commercial reactor due to its high fusion cross section (σ_f) at relatively low plasma temperatures (figure 1). Additionally, due to the significant mass difference between the emitted particles, 80% of the output energy (14.1MeV) is carried away by the neutron (Heckrotte and Hiskes, 1971). Such a high energy neutron is very desirable from energy production standpoint since it allows most of the energy produced by the reaction to be extracted from the plasma.

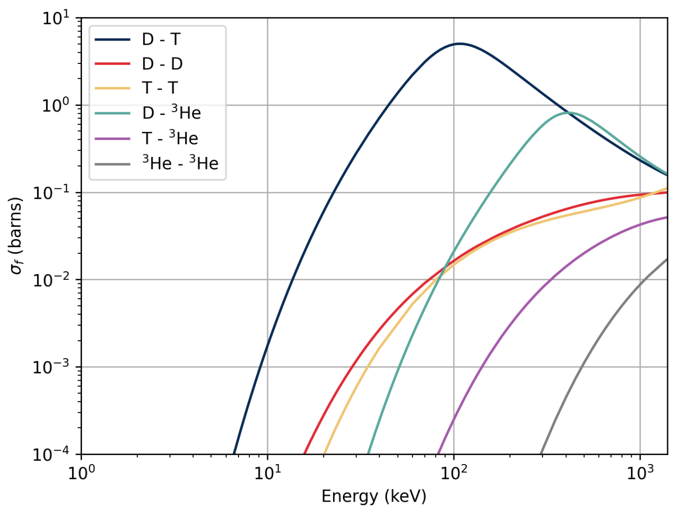

Figure 1: A graph showing the fusion cross sections (σ_f) of several fusion reactions for a range of energies. The data for this graph was collated by the Brookhaven National Laboratory for the Evaluated Nuclea Data File (ENDF B/VIII.1) (Brown et al., 2018).

1.1.2 Tritium in Fusion

Unfortunately, while deuterium is stable with a natural abundance of 0.0156%, tritium is not found in nature since it only has a half-life of 12.32 years. Thus, tritium must be produced. Most of the world's current tritium supply is generated in heavy water moderated fission reactors when neutrons are absorbed by the deuterium in the water. However, this process is costly and requires dedicated Tritium Removal Facilities (TRF's) of which only two exist (in Canada and South Korea) with another under construction (in Romania) (Pearson, Antoniazzi and Nuttall, 2018). In short, global tritium stockpiles are strained even without demand from fusion. If the International Thermonuclear Experimental Reactor (ITER) meets its operational targets, tritium stockpiles could run dry by 2049 (Kovari et al., 2018) . This has incentivised the design of an in-situ method of tritium production known as a breeder blanket. Initially envisaged as a method of creating fissile materials for fission reactors (Moir, 1982) (Maniscalco et al., 1984), a breeder blanket is a structure that produces desirable isotopes upon interacting with fusion neutrons. Crucially for us, neutrons interacting with lithium isotopes produce tritium. This is the basis of modern breeder blanket design and our project.

1.2 Breeder Blankets

Breeder blankets can be divided into two main categories, liquid breeders consisting of molten lithium compounds and solid breeders based on ceramic lithium oxides. The key differences between these solid and liquid breeders are in terms of thermal properties and tritium extraction (Konishi et al., 2017), qualities that have limited impact on our project but are of considerable importance to the design of practical breeder blankets. Comparitive research into practical blaket designs is becoming increasingly important as construction of functioning test reactors is expexted to begin in the 2040's and 50's. Strategy documents are already outlining conceptual designs for blankets on ITER's prospective DEMO (DEMOnstration Power Plant) reactors (Federici, Biel et al., 2017), the envisaged Chinese Fusion Engineering Test Reactor (CFETR) facility (Liu et al., 2022) and the Spherical Tokamak for Energy Production (STEP) planned for the UK (Lord et al., 2024), to name just a few.

1.2.1 Liquid Breeders

The simplest possible breeding blanket is one made of pure molten lithium. However, D-T fusion only produces one neutron and only one tritium atom is produced per lithium decay. Any neutron losses from this system would inevitably lead a tritium deficit. To overcome this, most blankets include a neutron multiplier, an isotope which releases additional neutrons. This will be discussed in more detail later in the report. A popular solution involves a liquid lithium-lead alloy (Tas et al., 1988) (Wu, 2011), typically chosen to be $Li_{17}Pb_{83}$. In addition to a breeder material, blankets must also include coolants. The primary candidates are Water-Cooled Lithium-Lead (WCLL) (Federici, Boccaccini et al., 2019) and Helium-Cooled Lithium-Lead (HCLL) blankets (Wu, 2011). Tritium extraction is also usually carried out by the coolant system. A variant of liquid breeder known as Dual-Cooled Lithium-Lead (DCLL) have been desiged so that the molten alloy also contributes towards the cooling, increasing the thermal efficiency of the blanket (Graham et al., 2022) (Fernández-Berceruelo et al., 2024).

Systems involving more chemically complex molten salts are seeing significant interest as a breeder for the Affordable Robust Compact (ARC) reactor developed by the Massachusetts Institute of Technology (MIT). The ARC project is dedicated to utilising an FLiBe ($2\,LiF$ - BeF_2) blanket, combining breeder and multiplier in a fluorine salt mixture with the goal of a reducing reactor size and increasing efficiency (Bocci et al., 2020) (Sorbom et al., 2015). In general liquid breeders are compact and chemically simple. However, liquid metal systems induce strong magnetohydrodynamic (MHD) currents that inhibit circulation (Morley, Malang and Kirillov, 2005).

1.2.2 Solid Breeders

Solid breeder designs have been proposed for several breeder projects, typically of mixture of a lithium ceramic and multiplier pebbles. Designs have coalesced around lithium orthosilicate (Li_4SiO_4) and lithium metatitanate (Li_2TiO_3) ceramics due to their tritium breeding and material properties. Beryllide ($Be_{12}Ti$) is the preferred moderator material for many solid breeder systems due to its resistance to oxidation. A Water-Cooled Pebble Bed (WCPB) is the primary focus of the CFETR project (Liu et al., 2022) (Lei et al., 2020). Meanwhile, ITER's solid breeder program is aimed at developing a Helium-Cooled Pebble Bed (HCPB) system for DEMO (Wang et al., 2019) (Federici, Boccaccini et al., 2019). The tritium breeding capability of a reactor can be significantly effected by the coolant systems. They require significant modification of the blanket structure and the coolants themselves have a small but noticable effect on neutron transport. A plethora of materially stable lithium ceramics and breeders are the key selling points of ceramic breeders (Hernández and Pereslavtsev, 2018). The packing factor of these pebble bed must be factored into their overall design (Gan, Kamlah and Reimann, 2010).

1.2.3 Contemporary Literature

With the construction of full scale test reactors and functional blanket modules anticipated within the next few decades a majority of the literature is focussed on a small selection of breeder materials and designs. These have been chosen not only for their tritium breeding properties but also for availability and structural reasons. More than 30 years ago some basic blanket designs and materials had already been narrowed down to fit these criteria (Shatalov et al., 1991). This allows our project to venture into less explored territory, studying compounds that have been discarded or overlooked due to their material properties, availability or cost. However, that is not to say that our project is unprecedented. The systematic modelling of TBR for different blankets and properties has been explored before in detail, usually as part of the design process for full scale test breeders (Hernández and Pereslavtsev, 2018), (Segantin, Testoni and Zucchetti, 2020) (Boullon, Jaboulay and Aubert, 2021). Of particular note is an unpublished paper by Goel, Aslam and Dua (2023) which offers us an example of a smaller, independent investigation closer to what we will be able to achieve with the resources available to us. These studies serve as useful sources of breeder materials and points of reference as the project progresses.

2 Theory

2.1 Neutron Interactions

2.1.1 Tritium Production

By irradiating lithium isotopes with neutrons, it is possible to produce tritium as seen in equations 2 and 3. A high tritium production reaction rate and relatively low neutron absorption makes lithium the preferred element for breeding.

$$^{6}\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + 4.8\,\mathrm{MeV} \tag{2}$$

$$^7\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + \mathrm{n} - 2.5\,\mathrm{MeV} \tag{3}$$

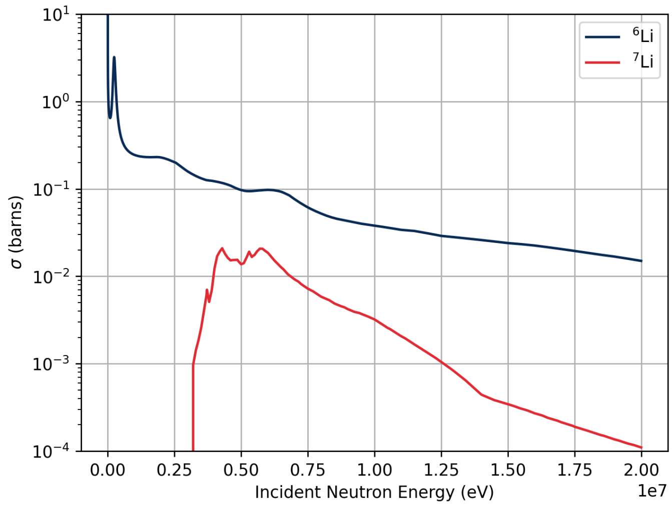

Figure 2: A graph of the tritium production cross sections ^{6}Li and ^{7}Li. Data gathered from ENDF B/VIII.1 (Brown et al., 2018).

The reaction cross section of ^{6}Li is several orders of magnitude larger than that of ^{7}Li, as can be seen in figure 2. Thus, despite the usefulness of extra neutron production during the ^{7}Li decay, ^{6}Li is the only realistic candidate for practical tritium breeding. With a natural abundance of only 1.9–7.8%, ^{6}Li must be enriched to achieve practical tritium production rates (NNDC, 2024). Lithium has the potential to form a wide variety of solid and liquid compounds, providing many possible materials with which to construct a functional tritium breeding blanket.

2.1.2 Neutron Multiplication

Charged particle emission is not the only interaction plays a key role in breeder blanket design. Neutron multiplication or 'spallation' is the process by which an incident neutron causes the emission of another. Thus, by including a multiplier within the structure the tritium production is no longer reliant only on fusion neutrons. Lead (equation 4) and beryllium (equation 5) are considered to be the best multiplier materials, once again due their high reaction cross sections (Tosti and Ghirelli, 2013).

$$^{208}\mathrm{Pb} + \mathrm{n} \longrightarrow {}^{207m}\mathrm{Pb} + 2\,\mathrm{n} - 7.0\,\mathrm{MeV} \tag{4}$$

$$^9\mathrm{Be} + \mathrm{n} \longrightarrow {}^8\mathrm{Be} + 2\,\mathrm{n} - 2.5\,\mathrm{MeV} \tag{5}$$

Figure 3 shows the microscopic neutron multiplication cross sections of lead and beryllium isotopes. Values for lead based multiplication are much higher than for beryllium but only for neutron energies greater than ~8Mev. Thus, the relative performance of these elements in a breeder blanket are highly dependent on neutron modulation and scattering in the rest of the structure. While ^{9}Be is the only beryllium isotope found in nature there are several stable lead isotopes. Fortunately, their cross sections are very similar, particularly at neutron energies

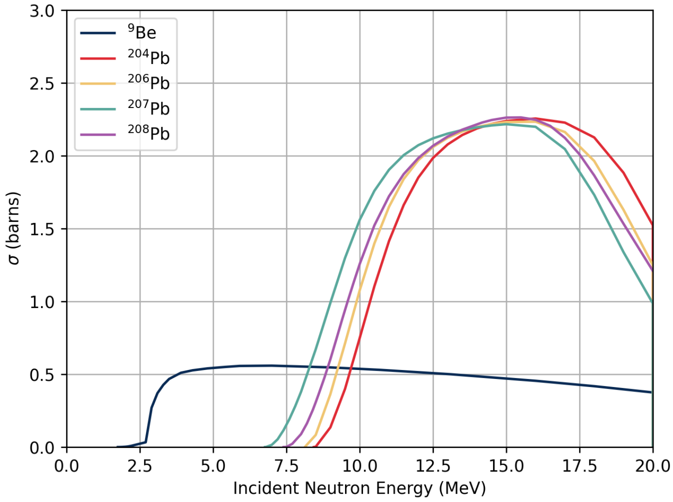

Figure 3: A graph of the neutron multiplication cross sections for stable lead and beryllium isotopes. Plotted using data from ENDF B/VIII.1 (Brown et al., 2018).

nearing 14.1MeV. The variation at lower energies is largely due to the nuclear shell structure of the isotopes.

2.2 Tritium Breeding Ratio

The primary metric by which we will be evaluating various blanket structures is the Tritium Breeding Ratio (TBR). It is the ratio between the amount of tritium produced by the blanket to the amount consumed during fusion. Assuming that tritium is only produced by ^{6}Li and ^{7}Li and that it is only consumed by the fusion reaction we get the following equation.

$$\mathrm{TBR} = \frac{R_t(^6\mathrm{Li}) + R_t(^7\mathrm{Li})}{R_{\mathrm{DT}}} \tag{6}$$

Due to neutron losses, structural vacancies and tritium leakage a TBR of 1 is typically not enough to ensure equilibrium in a realistic fusion reactor. In general, a value of between 1.05 and 1.1 is the minimum required for tritium self-sufficiency (Tanabe, 2017).

2.3 Neutron Transport

To obtain a TBR for a breeder structure we must calculate the tritium breeding reaction rates. The first step is modelling the paths and interactions of neutrons. Scattering, absorption, induced fission and radiative capture are a few of the processes that take place between neutrons and materials. Each of these interactions can be quantified by a macroscopic cross section,

$$\Sigma_x = n_i \sigma_x, \tag{7}$$

composed of the atomic density of the material (n_i) and the microscopic cross section (σ_x). The angular density of neutrons with energy E at time t in a direction $\vec{\Omega}(\theta, \varphi)$ at distance $\vec{r}$ from the origin is expressed as $N(\vec{r}, E, \vec{\Omega}, t)$. Multiplying density by neutron velocity ($|\vec{v}|$) leads to an expression for neutron flux.

$$\Phi(\vec{r}, E, \vec{\Omega}, t) = |\vec{v}| N(\vec{r}, E, \vec{\Omega}, t) \tag{8}$$

Neutron flux is a key component in the steady state neutron transport equation. It is a variation on the Boltzmann equation used to study thermodynamic statistical behaviour.

$$\vec{\Omega} \cdot \nabla\Phi(\vec{r}, E, \vec{\Omega}) + \Sigma_t \Phi(\vec{r}, E, \vec{\Omega}) = \int \Sigma_s(\vec{r}; E' \to E, \vec{\Omega}' \to \vec{\Omega}) \Phi(\vec{r}, E, \vec{\Omega}) dE' d\Sigma' + S(\vec{r}, E, \vec{\Omega}) \tag{9}$$

Each term is an expression for a contribution to the overall neutron balance at a specific point $\vec{r}$ with solid angle $\vec{\Omega}$ and energy E. Terms on the left describe neutron losses due to outward neutron flux and a combination of scattering and absorption. Those on the right characterise neutron gains resulting from source neutrons and backscatter. Accurately simulating the neutronics of breeder materials requires that we solve this equation computationally (Tanabe, 2017) (Kuridan, 2023).

2.4 Monte Carlo Simulations

Solutions to the neutron transport equation can be calculated using a Monte Carlo (MC) simulation. Developed by Stanislaw Ulam and John von Neumann in the 1940's, Monte Carlo simulations provide a useful process for modelling a wide variety of physical systems (Summerscales, 2023). Named for the seaside municipality and its famous casinos, MC calculations are probabilistic. By applying statistical sampling methods, they can be used to solve a variety of mathematical and scientific problems. With the advent of modern computing, it is now possible to simulate very complex and multivariate processes very quickly. Nuclear physics, finance and traffic management are just a few of the many fields that have benefitted from MC simulations.

The Monte Carlo process is based on the principle that a random sample of a population tends to retain the properties of the whole. Instead of precisely modelling each and every object a random selection can produce the same solution. By defining dependent and independent variables and assigning probabilities to the events under consideration a model of possible results can be constructed (Kalos and Whitlock, 2008) (Kuridan, 2023). Monte Carlo neutronics models create a sample of neutrons and then simulate a 'random walk' with a length determined by

$$l = -\frac{\ln(\xi)}{\Sigma_t}. \tag{10}$$

By cycling through a large set of pseudorandom numbers (ξ) very accurate neutron flux and TBR solutions can be produced. As the number of iterations increases so does the accuracy of the model (Romano et al., 2015).

3 Methodology

3.1 OpenMC

Monte Carlo codes are regularly used to model the neutronics of breeder designs planned for full scale international ventures (Wu, 2011) (Zheng and Todd, 2015). This project uses

OpenMC, a Monte Carlo code designed specifically for neutron transport in constructive solid geometry (CSG) environments. OpenMC pulls values from the ENDF databases to create a free gas approximation of neutron kinematics. Neutron paths and velocities are sampled from a Maxwellian distribution. So called 'tallies' of key parameters such as neutron flux, energy deposition and most importantly TBR can be produced. Careful filtering allowed us to calculate tallies for entire structures (Romano et al., 2015). It is by this process that the tritium production capabilities of a breeder blanket can be precisely measured. OpenMC has been shown to demonstrate very close agreement with other neutron transport codes when modelling neutron flux spectra, power distribution and TBR (Bae, Peterson and Shimwell, 2022) (Valentine et al., 2022). Our choice of programme is further justified by the examples of its use in studying the TBR of very complex blanket structures (Fradera et al., 2021) (Segantin, Testoni and Zucchetti, 2020).

3.2 Paramak

A significant challenge of this project was modelling the structure of a fusion reactor. OpenMC has the capability to produce simple simulation geometries using Constructive Solid Geometry. Unfortunately, the models are time consuming to create, especially for more complex reactor designs. A detailed study of breeder blankets requires something more specialised. Paramak is the solution, a CAD framework containing a suite of ready-built reactor designs (Shimwell et al., 2021). Instead of programming a simplistic model from scratch a fully realised Tokamak structure can be imported from a set library. This allows us to perform simulations much faster and with confidence that our results are representative of practical systems.

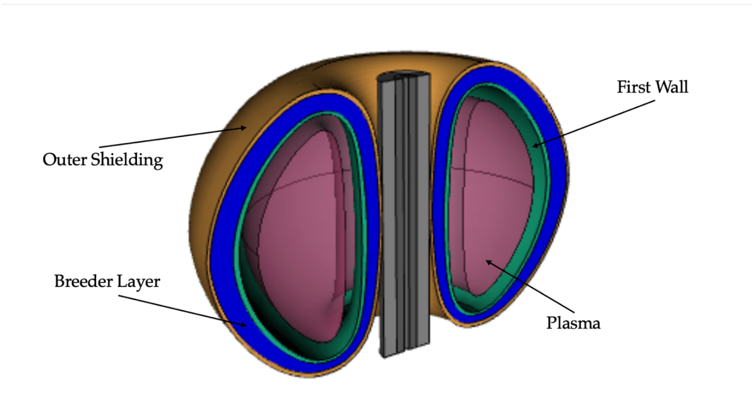

Figure 4: A visual representation of the Tokamak design used in our simulations. This design was imported from the Paramak library of fusion reactor structures created by Shimwell et al. (2021). Key structures such as the plasma and the breeder blanket are indicated in the figure.

Figure 4 depicts the basic Tokamak model we used in our simulations. It is a simple toroidal structure containing a plasma surrounded by a first wall, a breeder layer and an outer wall. One of the key benefits of Paramak models is the ease by which the specifications can be modified. The thickeness of the reactor components or the triangularity of the plasma vessel are two obvious examples. As a result, we were able to study the TBR behaviour for a sweep of parameters.

3.3 Simulation Parameters and Implementation

3.3.1 Neutron Source Parameters

Before embarking on our investigations of different breeding blankets we had to ensure that we had confidence in the results our simulation would produce. It was important that we defined a set of base attributes that would remain constant while individual parameters were varied. Drawing from a wide variety of sources we attempted to guarantee that our blanket had dimensions and compositions that would be expected in a realistic fusion reactor. Modelling our neutron source was the first step in this process. We decided to employ an isotropic, D-T, ring source placed at the approximate plasma radius. A Muir distribution (Muir, 1974) centred on 14.1MeV was used to define the neutron energies. This is a result of the thermal excitation of the plasma ions whose temperature was set at 20keV (Wandelt, 2018). For simplicity we have not adopted a plasma based neutron source. Thus, the attributers of the plasma layer in the Paramak model do not affect the source parameters, only the neutron transport once they have been produced. Our plasma characteristics remained constant throughout the simulation process.

3.3.2 Blanket Materials and Composition

For our standard breeder material we chose $Li_{17}Pb_{83}$ for its chemical simplicity and popularity within the available literature (Malang et al., 1991) (Wu, 2011) (Fernández-Berceruelo et al., 2024). This specific combination is also eutectic, which means it has the lowest melting point of all possible lithium-lead mixtures, a vital property in molten metal systems (Mas de les Valls et al., 2008). ^6Li enrichments varied wildly between different sources ranging from as low as 10% to as high as 90% (Segantin, Testoni and Zucchetti, 2020) (Boullon, Jaboulay and Aubert, 2021). After some consideration we adopted a value of 60% used by Hernández and Pereslavtsev, 2018, representing an 'average' enrichment value. In addition to the breeder material we also require a coolant within the blanket structure. As discussed previously, the vast majority of blanket designs plan to use either water (Hirose et al., 2024) or helium (Chen et al., 2015) however, a small handful of projects are exploring a nitrogen based cooling system (Fradera et al., 2021). We opted for a water based cooling system occupying 33% of the breeder volume. To ensure that our simulation did not balloon in complexity we decided to combine the breeder and coolant into one homogenous layer.

3.3.3 Reactor and Blanket Dimensions

While we had used Paramak to create a pre-built tokamak design (see figure 4) we still needed to specify the dimensions and composition of the structure. The most important of these were the first wall and blanket thicknesses. As the name suggests, the first wall is the first layer separating the plasma from the rest of the reactor. As outlined in King et al., 2022, first wall materials require low neutron interaction cross sections and strong physical properties. The Zr-4 alloy (descibed in the same paper) best suited our needs due to the low neutron interaction cross section if zirconium. The first wall was set to a thickness of 25mm based on the available literature (Boullon, Jaboulay and Aubert, 2021) (Lei et al., 2020). Based on designs in Lei et al., 2020 and Liu et al., 2022 the breeder blanket depth was set at 500mm. The shape of the torus meant that the blanket layer was not entirely uniform, therefore the 500mm depth is an average over the whole structure. The final parameter of note was the triangularity. As the name suggests, triangularity is a description of how triangular the inner cavity is. Modifying this property significantly distorts the shape of our plasma and its positioning in relation to the breeder layer. Throughout our simulations used a value of 0.55 which had been pre-set in the Paramak model.

4 Results

4.1 Blanket Material Results

4.1.1 Liquid Breeder Materials

We began our investigation by exploring the TBR ratio for a wide variety of breeder materials. Materials and their properties were derived from similar TBR investigations in the literature (Hernández and Pereslavtsev, 2018) (Segantin, Testoni and Zucchetti, 2020) (Boullon, Jaboulay and Aubert, 2021). Table 1 lists the data for liquid breeders with the parameters previously outlined. Many of these are mixtures so the molar ratio of compounds is noted alongside the lithium content and density.

Liquid Breeder TBR					
Material	Ratio (molar %)	Li (molar %)	ρ (g/cm^3)	TBR	Heating (MeV)
$Li_{17}Pb_{83}$	-	17.00	11.00	1.61	11.54
$LiF-BeF_2$	2:1	28.57	2.04	1.15	12.92
$LiF-BeF_2$*	1:1	20.00	2.06	1.16	12.97
$LiF-PbF_2$	2:3	15.38	3.55	1.23	11.63
LiF	-	50.00	2.64	1.21	13.49
Li	-	100	0.51	1.17	13.60
Li_3N	-	75.00	1.30	1.13	13.33
$LiF-BeF_2-NaF$	1:1:1	14.29	2.15	1.09	12.42
$LiF-NaF-ZrF_4$	55:22:23	20.45	2.72	1.07	11.59
$LiF-LiI_2$	83.5:16.5	50.00	3.68	1.02	10.96
$LiF-LiBr-NaF$	14:79:7	45.69	3.20	0.98	15.57
$LiF-LiBr-NaBr$	20:73:7	46.50	3.16	0.97	15.55
$LiCl-PbCl_2$	1:2	12.50	4.50	0.96	9.87
$LiCl-NaCl$	72:28	36.00	1.52	0.91	11.04
$LiF-NaF-KF$	46.5:11.5:42	20.17	2.02	0.90	11.33
$LiCl-BeCl_2$	1:1	20.00	1.52	0.88	10.90
$LiCl-KCl$	7:3	35.00	1.51	0.85	10.72
$LiCl-KCl$*	52:48	26.00	1.51	0.80	10.37

Table 1: A table detailing the TBR and heating values of several liquid breeder materials. The chemical formulae of these compounds are listed along with their chemical ratio (for mixtures), lithium content and density. Materials above the dashed line are those that have been considered for practical breeder blankets. Uncertainties for the TBR and heating were less than 1%.

Along with TBR, the neutron heating was also measured. In this case heating is defined as the energy deposited in the reactor structure per source neutron. Since we plan to use this reactor to produce energy a higher heating is generally preferred however it TBR remained the primary metric by which we would evaluate blanket suitability. Materials listed above the dashed line are those currently under consideration for future use in fully operating fusion reactors. It is important to note that, due to a high particle number and simulation batches, the uncertainty on the tabulated values is incredibly low, usually less than 1%. Of the liquid breeder materials eutectic $Li_{17}Pb_{83}$ had by far the highest measured TBR for the simulation parameters outlined in the methodology.

4.1.2 Solid Breeder Materials

A comparable study was carried out on the list of solid breeder materials as shown in table 2. Beforehand some changes had to be made to the model attributes. Solid breeders usually come in the form of a bed consisting of distinct breeder and multiplier pebbles. For these simulations a $Be_{12}Ti$ multiplier was used at a breeder to multiplier ratio of 1:3 (Hernández and Pereslavtsev, 2018). Additionally, solid spheres can only be packed to fit 62% of the allotted space (Gan, Kamlah and Reimann, 2010), leading to lower effective densities than those quoted in table 2. Apart from these modifications all other parameters remained constant. As with the liquid breeders, both TBR and heating were recorded.

Solid Breeder TBR				
Material	Li (molar %)	ρ (g/cm^3)	TBR	Heating (MeV)
Li_4SiO_4	44.44	2.40	1.47	14.88
Li_2TiO_3	33.33	3.43	1.46	14.63
Li_8PbO_6	53.33	4.28	1.51	15.01
Li_8CeO_6	53.33	3.25	1.51	15.29
Li_8ZrO_6	53.33	2.98	1.50	14.99
Li_8SnO_6	53.33	3.41	1.49	15.18
$Li_6Zr_2O_7$	40.00	3.56	1.49	14.79
Li_8GeO_6	53.33	2.64	1.49	14.97
Li_5FeO_4	50.00	2.64	1.48	14.91
Li_5AlO_4	50.00	2.25	1.48	15.00
Li_8SiO_6	53.33	2.20	1.48	15.03
Li_6ZnO_4	54.55	2.86	1.48	14.95
Li_4TiO_4	44.44	2.57	1.48	14.89
Li_6MnO_4	54.55	2.50	1.48	14.91
Li_4GeO_4	44.44	3.16	1.48	14.79
Li_2O	66.66	2.10	1.47	14.92
Li_2SiO_3	33.3	2.53	1.45	14.75
Li_2MnO_2	40.00	3.90	1.43	14.42
Li_8CoO_6	53.33	2.47	1.43	14.69
Li_6CoO_4	54.55	2.77	1.41	14.58

Table 2: This table notes the TBR and heating of common solid breeder ceramics along with their lithium content and density. All results were produced with a $Be_{12}Ti$ multiplier. Similarly to the liquid breeder values, uncertainties were of the order of 1%.

4.1.3 Blanket Material Composition

Our next step was to study the effects of modifying the composition of the breeder material. We began by varying the Li:Pb ratio within the lithium-lead alloy. As exhibited in figure 5, the TBR rises with increasing lead fraction, peaking at $Li_{45}Pb_{55}$ before declining steadily. The graph also displays the TBR for the eutectic (and hence denser) $Li_{17}Pb_{83}$ mixture.

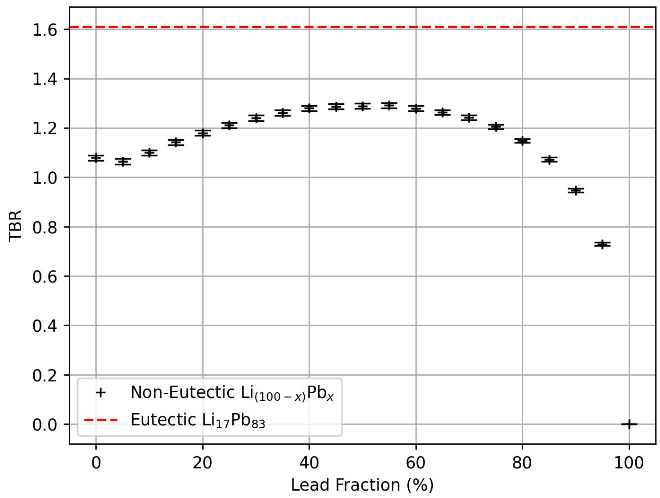

Figure 5: A plot of the TBR for a series of non-eutectic lithium-lead mixtures in comporison with eutectic $Li_{17}Pb_{83}$.

We can also modify the ^{6}Li enrichment within the mixture. As demonstrated by figure 6, increasing ^{6}Li enrichment increases TBR up to around 40% before reaching a plateau. As it climbs past 80% the TBR begins to decline slightly.

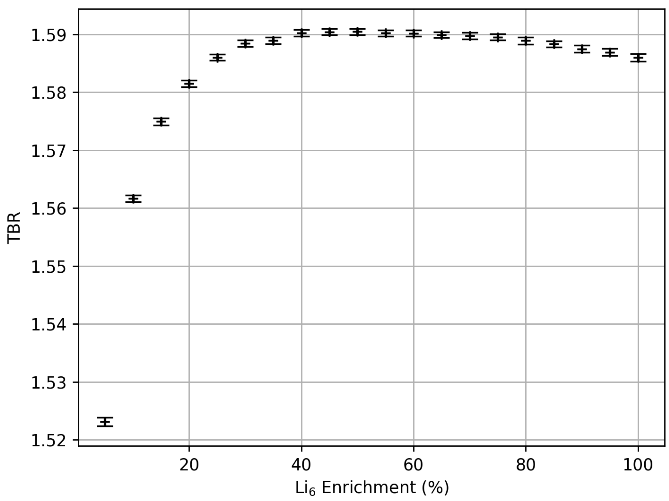

Figure 6: This graph shows the change in TBR for a range of ^{6}Li enrichment values. Uncertainties are small but present.

4.1.4 Coolant Fraction

So far, we have assumed that 33% of the breeder layer is filled with a water coolant in a homogenous mixture. Sweeping for a range of coolant volume fractions results in the graph on the left of figure 7. Three coolants were simulated, water, helium and nitrogen. As expected, increasing proportions of helium and nitrogen led to reduction in the TBR. In contrast the TBR increased for small fractions of water and undertook a steadier decline at higher proportions.

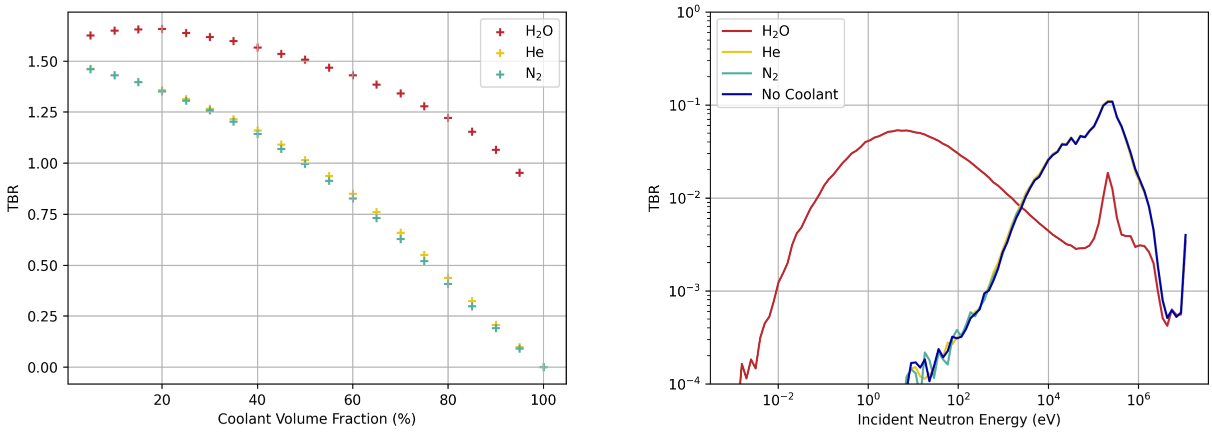

Figure 7: Two graphs exhibiting the effects of water, helium and nitrogen coolants on TBR. On the left, TBR is shown for at varying fractions of coolant in the breeder layer. The graph on the right depicts the contribution to TBR for a sweep of neutron energies. Uncertainties are present but small.

The rightmost graph of figure 7 displays the tritium breeding contribution of different neutron energies for each coolant type and for a blanket with no coolant. Neutron energies in the water-cooled blanket are significantly lower than the others which overlap almost entirely.

4.2 Blanket Geometry Results

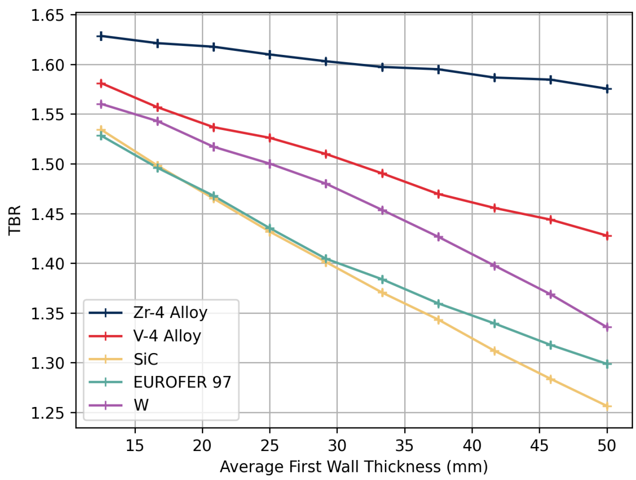

Figure 8: A graph of the change in TBR for the thicknesses of several common first wall materials.

In addition to modifying the content of the breeder blanket we can also vary the dimensions of the reactor. It is easy to imagine the effect that changing the effect that changing the thicknesses of key components will have on neutron attenuation and tritium breeding. We began by studying the first structure that neutrons with encounter after leaving the plasma, the firs wall. As mentioned previously the first wall must be structurally resistant enough to shield key componenets from the high plasma potential and allow neutrons to pass through so that tritium production can take place. Figure 8 shows how TBR declines with increasing thickness for several candidate first wall materials whose compositions are detailed in (King et al., 2022). The Zirconium-4 alloy is by far the best in terms of tritium breeding performance.

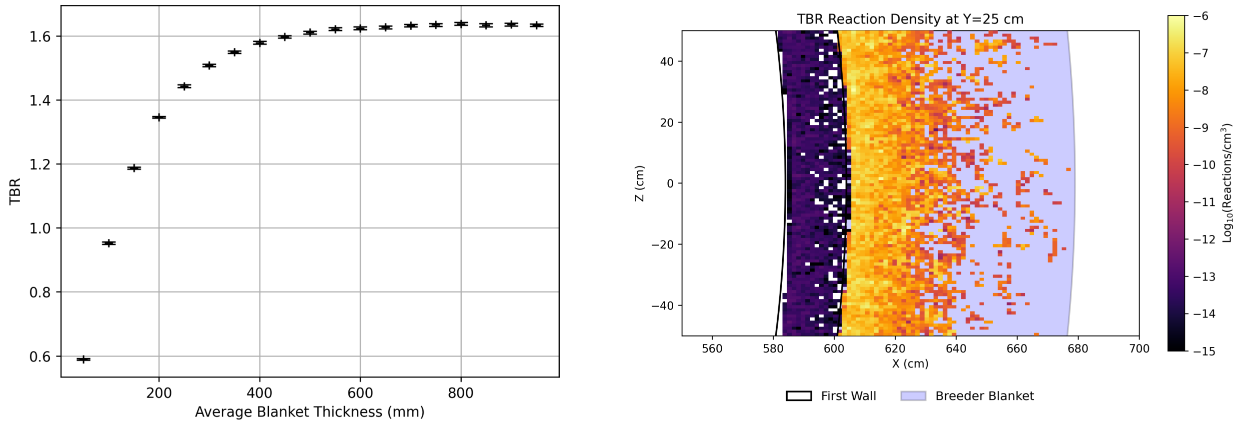

Figure 9: This figure contains two graphs. On the left, TBR is plotted for increasing blanket thickness. On the right, tritium breeding reaction density is shown for a 2D slice of the breeder.

After passing through the first wall the neutrons reach the breeder module. This layer contains the materials responsible for tritium production, so naturally it's dimensions will have a significant effect on TBR. The graph on the left of figure 9 demonstrates this clearly. TBR increases very quickly with blanket thickness up to a value of 200mm where it begins to slow, levelling off at around 500-600mm. The tritium breeding reaction density within the first wall and breeder blanket is plotted in the graph on the right of figure 9. It shows that travelling radially outwards from the breeder, the reaction density begins to decline very quickly beyond around 200mm. A vast majority of neutrons react with the breeder very close to the surface.

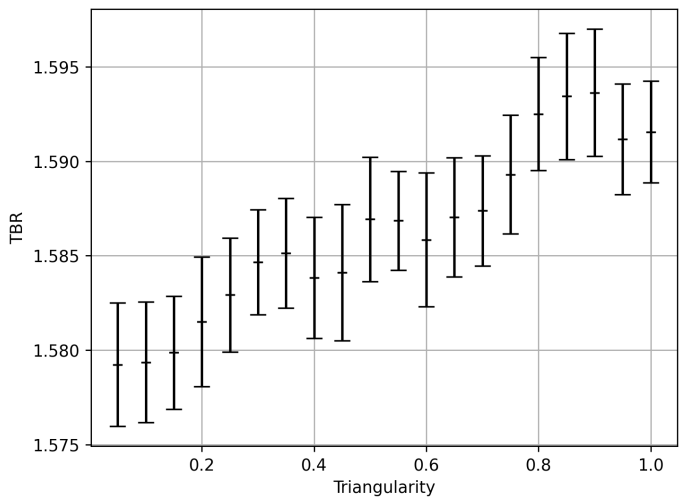

Figure 10: A plot of the TBR against the triangularity of the reactor torus.

Another possible modification to the tokamak design is the triangularity of the torus. Figure 10 shows the effect that changing it had on the TBR. The very large uncertainties present in this plot are result of the meagre change in TBR. While there is a slight increase as triangularity increases, the disproportionate uncertainties cast doubt on any correlation.

5 Discussion

5.1 Blanket Materials

Our report has outlined our investigation into the tritium production capabilities of fusion breeder blankets. Using a simplified reactor model, we carried out investigations of several key parameters to confirm the trends observed in literature and to characterise an 'optimal' tritium breeder. Perhaps the most obvious is that eutectic $Li_{17}Pb_{83}$ has by far the highest TBR of the blanket materials we investigated. This is a result of its high density and more importantly, its significant neutron multiplier fraction. We can be relatively confident of the validity of this outcome since it is replicated across several other tritium production simulations (Goel, Aslam and Dua, 2023) (Segantin, Testoni and Zucchetti, 2020). Table 1 demonstrates that for many of the liquid compounds we investigated they lack a dedicated neutron multiplier element, resulting in a neutron deficit and the plethora of TBR values less than equilibrium. While we may have expected a higher lithium content to result in more tritium production this is not the case. Clearly other factors such as neutron multiplication, scattering and particle density hold more sway. Apart from $Li_{17}Pb_{83}$, the liquids that performed best were the fluorine based molten salts. These had much better tritium production capabilities than the chlorine or bromine salts. Fluorine has much a lower neutron absorption cross section than both, resulting in more neutrons available for tritium breeding. Thus, while there has been some study into chlorine salts in breeders (Bohm and Lindley, 2023) fluorine remains the preferred option for molten salt blankets.

The chemical similarity the solid breeders produces TBR results that are much closer to each other than those of the liquids. In general, the TBR was also higher for solids. This is undoubtedly due to the presence of the $Be_{12}Ti$ multiplier pebbles in the solid blankets, guaranteeing that there are plenty of available neutrons. Compounds containing the most lithium atoms usually produced higher TBR values, following the trend shown in Hernández and Pereslavtsev, 2018. However, there are a few key exceptions. Namely, the cobalt ceramics which, due to the high absorption cross section of cobalt, have the lowest tritium breeding performances. The results of the table 2 show that while $Li_{17}Pb_{83}$ is the best material overall, solid breeders are more consistently effective tritium breeders. We are justified in using $Li_{17}Pb_{83}$ for our simulation however, in a more functional breeder design, material and thermodynamic constraints may bias towards a solid blanket.

In addition to TBR we also measured the heating energy produced in the structure per source neutron. A higher heating indicates that more energy is being extracted from the source neutrons, a key characteristic for energy producing fusion reactors. In general, our data shows an increased heating correlated with a higher TBR. A vast majority of tritium production is a result of the exothermic ^{6}Li reaction shown in equation 2. Thus, more tritium production reactions lead to more energy produced per neutron. There are a few key outliers. Bromine salts have much higher than expected heating arising from the very large absorption cross section of bromine isotopes. On the other hand, compounds containing lead tend to have lower heating due to the very endothermic neutron multiplication reaction (equation 4). This is another factor that, in a fully functioning reactor where energy production is crucial, may count against a $Li_{17}Pb_{83}$ breeder.

An important property of $Li_{17}Pb_{83}$ is that it is a eutectic mixture. This means it has a melting point 100 C lower than that of pure Pb a very useful feature for molten metal breeders (Mas de les Valls et al., 2008). More importantly for our study is that it has a 15% higher density than we would expect for those proportions. This is the reason for the significant discrepancy in results between the TBR of eutectic $Li_{17}Pb_{83}$ and of the non-eutectic lithium-lead fractions plotted in figure 5. Thus, while the non-eutectic data would initially suggest that we opt for a $Li_{45}Pb_{55}$ mixture, the eutectic alloy is clearly superior, both from a materials and a TBR perspective.

Increasing the ^{6}Li enrichment from natural abundance results is a reliable way of improving TBR. The tritium production cross sections plotted in figure 2 show that the ^{6}Li reaction is much more probable, particularly at lower energies. Figure 6 begins to show this trend up to an enrichment of 40%. Beyond this value the TBR steadies and then declines slightly. We have reached the point of neutron saturation, where the addition of more ^{6}Li atoms isotopes with not lead to more tritium since there are no neutrons to interact with. As the enrichment rises further the TBR begins to decline. Equation 3 shows that the ^{7}Li tritium production reaction also releases neutrons. Thus, when ^{7}Li becomes scarce the number of neutrons in the system is decreased slightly, lowering the overall TBR. These results are encouraging; they indicate that for our simulation a somewhat lower enrichment of 40% would not only be acceptable but preferential. Lithium-6 enrichment is a difficult, costly and environmentally hazardous process (Giegerich et al., 2019), so minimising the ^{6}Li content is key.

As those familiar with nuclear fission reactors will be aware, water is a very effective neutron moderator. Thus, the neutrons interacting with blankets containing water will have much lower energies. This explains the higher TBR measured for water-cooled blankets in the leftmost graph of figure 7. The tritium production cross section of ^{6}Li increases logarithmically at low incident neutron energies. Plotting the tritium production contributions of a range of neutron energies results in the graph on the right of figure 7. For a water-cooled blanket the TBR contribution is shifted significantly towards neutron energies of less than 1eV. Meanwhile the spectra for blankets containing helium and nitrogen are virtually identical to the results with no coolant at all. Not only does this graph clearly demonstrate the neutron moderation properties of water but it shows that helium and nitrogen have essentially no effect on neutron energy. It is obvious that from a tritium production point of view, a water coolant is the optimal choice. Combined with its superior cooling properties it is no surprise that water is so widely used in breeder designs. However, the violent chemical reaction between water and lithium is a key consideration and unlike in our homogenous design, real systems take great care to keep these elements separated.

Combining these results we can safely say that for our simulation model the best performing tritium breeder was a water-cooled lithium-lead (WCLL) blanket. WCLL blankets are one of the more popular blanket configurations currently being considered by ITER for their DEMO reactor. The advantages of this setup is its simplicity. Water is a cheap and efficient coolant, $Li_{17}Pb_{83}$ is a simple and powerful breeder. However, as has been touched on previously this system suffers from several material challanges that were not considered in our investigaiton of TBR. The most concerning of these being the MHD effects of liquid metals, the dangers invloved when water and lithium are in close proximity and potential difficulties in extracting tritium from a liquid breeder.

5.2 Blanket Geometry

As shown in figure 8, a first wall consisting of the Zirconium-4 alloy has the best tritium TBR. Zirconium has a lower neutron interaction cross section than the other materials that have been considered as first wall candidates. This was suitable for our simulation since we prioritised tritium breeding above all other considerations. However, the first wall must also thermally shield the plasma and withstand a variety of stress forces. Consequently, the most common first wall material in literature is EUROFER 97, a Reduced Activation Ferritic-Martensitic (RAFM) steel designed specifically for the harsh environment of an operational fusion reactor. Most blanket design use EUROFER or a similiar high-strength, low-activation steel for their structural components. Clearly or focus on TBR as our primary method of evaluation does not necessarily result in the most ideal design.

Once they have travelled through the first wall the neutrons then reach the breeder blanket. We would expect neutrons to be attenuated by the material and that most of the tritium breeding reactions occur close to the blanket surface. This hypothesis is supported by the graphs in figure 9. The graph on the left shows the TBR increasing with blanket thickness up to around 200mm, before beginning to plateau. There is still some increase past this point however by 500mm it is negligible. This is mirrored in the rightmost graph that depicts the tritium breeding reaction density decreasing sharply beyond 200mm. Using the mean free path of Pb ($\sim$55mm) which constitutes the majority of the blanket we find that 97.4% of incident neutrons will have interacted in the first 200mm and 99.9% in the first 500mm. Any increase beyond this point would not measurably increase the TBR and would be wasteful and inefficient. The levelling out corresponds with the available literature, where most detailed blanket designs have modules 500mm deep, and with our base simulation which uses the same value.

In addition to the TBR reaction density in the first wall, the heat map in figure 9 also shows a small contribution in the first wall. It is a result of neutron interaction with zirconium.

$$^{90}\mathrm{Zr} + \mathrm{n} \longrightarrow {}^{88}\mathrm{Y} + \mathrm{t} - 11.4\,\mathrm{MeV} \tag{11}$$

Fortunately, this interaction only occurs for neutron energies above 14.5MeV, which only applies to unattenuated source neutrons at the higher end of the Muir distribution. Combined with a very low reaction cross section, it is safe to say that the contribution of this reaction to the total TBR is negligible. Thus, we can rule out the possibility TBR contributions from this reaction and confirm that zirconium would make a very poor breeder material.

Our final modification to the structure was performed by varying the triangularity, shown in figure 10. The graph reveals that this had very little effect on tritium production. The increase in TBR is miniscule and very large errors mean that any conclusions drawn from this data would be suspect. When we consider the parameters of our simulation this is understandable. Using a fixed ring source means that any changes to the plasma shape do not modify the source parameters. While this would modify the blanket shape somewhat it is evidently not enough to cause a significant change in TBR. A more realistic plasma source would likely produce a more conclusive result and offers an interesting avenue to explore in future simulations.

5.3 Model Limitations and Improvements

Using the characteristic defined in our base model, we have systematically investigated a variety blanket parameters to maximise the TBR. We have also managed to reproduce several key trends that have been identified and built into modern breeder designs. Despite these successes, the weaknesses of our model are clear. Our simulation is a great deal simpler than those used in the design and planning of full scale breeder projects. While this may have been suitable for a preliminary investigation it undoubtedly affected the accuracy of our results. To begin with, our neutron source was incredibly simplistic, failing to account for the effects of plasma shape and composition. The tokamak model we imported from Paramak was a useful design to start with but will need significant improvement if we want to compare our results to real fusion reactors. With these weaknesses in mind next steps will likely be to introduce more complexity into our simulation. We hope to employ more advanced Paramak models and potentially make our own changes to the structure. Introducing heterogeneity will be a key part of this process, particularly in separating the breeder material and the coolant. The effect on neutron moderation is an avenue we are very interested in exploring. Utilising a more realistic, plasma based neutron source is another route we intend to explore. This will allow us to more effectively study the effects of varying plasma characteristics and of modifying the tokamak structure. If the scope of our project allows, we may also be able to use this in combination with the more detailed structural models to simulate specific reactors and how this effects the optimal TBR parameters.

6 Conclusion

In summary, by carrying out a series of neutronics simulations we have produced TBR results for a variety of breeder blanket characteristics. These findings have allowed us to construct an optimal tritium production configuration for our simple, homogenous breeder. We have demonstrated that a water-cooled, lithium-lead (WCLL) breeder was the most effective tritium breeder. The high TBR values of lithium-lead found in literature and the prevalence of WCLL designs support the validity of our simulation and results. Furthermore our results for parameter sweeps of enrichment, neutron energy and blanket thickness showed good agreement with our prior expectations and with similar TBR investigations. We also identified potential subject for future research. Improvements to the fidelity of the breeder structure are an investigative avenue we hope to explore in significant depth.

References

Azizov, E. A. (2012). 'Tokamaks: from A D Sakharov to the present (the 60-year history of tokamaks)'. *Phys.-Usp.*, **55** (2), pp. 190–203. DOI: 10.3367/UFNe.0182.201202j.0202.

Bae, J. W., Peterson, E. and Shimwell, J. (2022). 'ARC reactor neutronics multi-code validation*'. *Nucl. Fusion*, **62** (6), pp. 1–17. DOI: 10.1088/1741-4326/ac5450.

Bocci, B. et al. (2020). 'ARC reactor materials: Activation analysis and optimization'. *Fusion Engineering and Design*, **154**, pp. 1–8. DOI: 10.1016/j.fusengdes.2020.111539.

Bohm, T. D. and Lindley, B. A. (2023). 'Initial Neutronics Investigation of a Chlorine Salt–Based Breeder Blanket'. *Fusion Science and Technology*, **79** (8), pp. 995–1007. DOI: 10.1080/15361055.2022.2136923.

Boullon, R., Jaboulay, J.-C. and Aubert, J. (2021). 'Molten salt breeding blanket: Investigations and proposals of pre-conceptual design options for testing in DEMO'. *Fusion Engineering and Design*, **171**, pp. 1–9. DOI: 10.1016/j.fusengdes.2021.112707.

Brown, D. et al. (2018). 'ENDF/B-VIII.0: The 8 th Major Release of the Nuclear Reaction Data Library with CIELO-project Cross Sections, New Standards and Thermal Scattering Data'. *Nuclear Data Sheets*, **148**, pp. 1–142. DOI: 10.1016/j.nds.2018.02.001.

Chen, H. et al. (2015). 'Conceptual design and analysis of the helium cooled solid breeder blanket for CFETR'. *Fusion Engineering and Design*. Proceedings of the 28th Symposium On Fusion Technology (SOFT-28), **96-97**, pp. 89–94. DOI: 10.1016/j.fusengdes.2015.02.045.

Federici, G., Biel, W. et al. (2017). 'European DEMO design strategy and consequences for materials'. *Nucl. Fusion*, **57** (9), pp. 1–26. DOI: 10.1088/1741-4326/57/9/092002.

Federici, G., Boccaccini, L. et al. (2019). 'An overview of the EU breeding blanket design strategy as an integral part of the DEMO design effort'. *Fusion Engineering and Design*, **141**, pp. 30–42. DOI: 10.1016/j.fusengdes.2019.01.141.

Fernández-Berceruelo, I. et al. (2024). 'Progress in design and experimental activities for the development of an advanced breeding blanket'. *Nucl. Fusion*, **64** (5), pp. 1–16. DOI: 10.1088/1741-4326/ad37ca.

Fradera, J. et al. (2021). 'Pre-conceptual design of an encapsulated breeder commercial blanket for the STEP fusion reactor'. *Fusion Engineering and Design*, **172**, pp. 1–15. DOI: 10.1016/j.fusengdes.2021.112909.

Gan, Y., Kamlah, M. and Reimann, J. (2010). 'Computer simulation of packing structure in pebble beds'. *Fusion Engineering and Design*, **85** (10), pp. 1782–1787. DOI: 10.1016/j.fusengdes.2010.05.042.

Giegerich, T. et al. (2019). 'Development of a viable route for lithium-6 supply of DEMO and future fusion power plants'. *Fusion Engineering and Design*, **149**, p. 111339. DOI: 10.1016/j.fusengdes.2019.111339.

Goel, V., Aslam, S. and Dua, S. (2023). *Optimization of Tritium Breeding Ratio in a DT and DD Submersion Tokamak Fusion Reactor*. DOI: 10.48550/arXiv.2310.00220.

Graham, G. et al. (2022). 'A Dual Coolant Lithium Lead Breeder Blanket for a Fusion Power Plant Systems Model'. *Journal of Nuclear Engineering*, **1**, pp. 1–8.

Heckrotte, W. and Hiskes, J. (1971). 'Some factors in the choice of D-D, D-T or D-3He mirror fusion power systems'. *Nucl. Fusion*, **11** (5), pp. 471–484. DOI: 10.1088/0029-5515/11/5/009.

Hernández, F. and Pereslavtsev, P. (2018). 'First principles review of options for tritium breeder and neutron multiplier materials for breeding blankets in fusion reactors'. *Fusion Engineering and Design*, **137**, pp. 243–256. DOI: 10.1016/j.fusengdes.2018.09.014.

Hirose, T. et al. (2024). 'Functional tests for water cooled ceramic breeder blanket system using full-scale mockups'. *Fusion Engineering and Design*, **200**, pp. 1–6. DOI: 10.1016/j.fusengdes.2024.114227.

Kalos, M. H. and Whitlock, P. A. (2008). *Monte Carlo Methods*. 1st ed. Wiley. DOI: 10.1002/9783527626212.

King, D. et al. (2022). 'High temperature zirconium alloys for fusion energy'. *Journal of Nuclear Materials*, **559**, pp. 1–27. DOI: 10.1016/j.jnucmat.2021.153431.

Konishi, S. et al. (2017). 'Functional materials for breeding blankets—status and developments'. *Nucl. Fusion*, **57** (9), pp. 1–19. DOI: 10.1088/1741-4326/aa7e4e.

Kovari, M. et al. (2018). 'Tritium resources available for fusion reactors'. *Nucl. Fusion*, **58** (2), pp. 1–10. DOI: 10.1088/1741-4326/aa9d25.

Kuridan, R. M. (2023). *Neutron Transport: Theory, Modeling, and Computations*. Graduate Texts in Physics. Cham: Springer Nature Switzerland. DOI: 10.1007/978-3-031-26932-5.

Lei, M. et al. (2020). 'Design and analysis of the equatorial inboard WCCB blanket module for CFETR'. *Fusion Engineering and Design*, **161**, pp. 1–9. DOI: 10.1016/j.fusengdes.2020.111889.

Liu, S. et al. (2022). 'Design of the Water-Cooled Ceramic Breeder blanket for CFETR'. *Fusion Engineering and Design*, **177**, pp. 1–8. DOI: 10.1016/j.fusengdes.2022.113059.

Lord, M. et al. (2024). 'Fusing together an outline design for sustained fuelling and tritium self-sufficiency'. *Phil. Trans. R. Soc. A.*, **382** (2280), pp. 1–15. DOI: 10.1098/rsta.2023.0410.

Malang, S. et al. (1991). 'Self-cooled blanket concepts using Pb-17Li as liquid breeder and coolant'. *Fusion Engineering and Design*, **14** (3), pp. 373–399. DOI: 10.1016/0920-3796(91)90020-Q.

Maniscalco, J. A. et al. (1984). 'The Fusion Breeder — an Early Application of Nuclear Fusion'. *Fusion Technology*, **6** (3), pp. 584–596. DOI: 10.13182/FST84-A23140.

Mas de les Valls, E. et al. (2008). 'Lead–lithium eutectic material database for nuclear fusion technology'. *Journal of Nuclear Materials*. Heavy Liquid Metal Cooled Reactors and Related Technologies, **376** (3), pp. 353–357. DOI: 10.1016/j.jnucmat.2008.02.016.

Moir, R. W. (1982). 'The fusion breeder'. *J Fusion Energ*, **2** (4), pp. 351–367. DOI: 10.1007/BF01063686.

Morley, N. B., Malang, S. and Kirillov, I. (2005). 'Thermofluid Magnetohydrodynamic Issues for Liquid Breeders'. *Fusion Science and Technology*, **47** (3), pp. 488–501. DOI: 10.13182/FST05-A733.

Muir, D. W. (1974). *Sensitivity of neutron multigroup cross sections to thermal broadening of the fusion peak*. CONF-740402-33 LA-UR-74-613. USA: Los Alamos Scientific Lab.

NNDC (2024). *Evaluated Nuclear Data File (ENDF)*.

Paris, M. W. and Chadwick, M. B. (2023). 'A lost detail in D–T fusion history'. *Physics Today*, **76** (10), pp. 10–11. DOI: `10.1063/PT.3.5317`.

Pearson, R. J., Antoniazzi, A. B. and Nuttall, W. J. (2018). 'Tritium supply and use: a key issue for the development of nuclear fusion energy'. *Fusion Engineering and Design*, **136**, pp. 1140–1148. DOI: `10.1016/j.fusengdes.2018.04.090`.

Romano, P. K. et al. (2015). 'OpenMC: A state-of-the-art Monte Carlo code for research and development'. *Annals of Nuclear Energy*, **82**, pp. 90–97. DOI: `10.1016/j.anucene.2014.07.048`.

Segantin, S., Testoni, R. and Zucchetti, M. (2020). 'Neutronic comparison of liquid breeders for ARC-like reactor blankets'. *Fusion Engineering and Design*, **160**, pp. 1–10. DOI: `10.1016/j.fusengdes.2020.112013`.

Shatalov, G. et al. (1991). 'Breeder and test blankets in ITER'. *Fusion Engineering and Design*, **16**, pp. 85–93. DOI: `10.1016/0920-3796(91)90185-S`.

Shimwell, J. et al. (2021). 'The Paramak: automated parametric geometry construction for fusion reactor designs.' *F1000Res*, **10**, p. 27. DOI: `10.12688/f1000research.28224.1`.

Sorbom, B. et al. (2015). 'ARC: A compact, high-field, fusion nuclear science facility and demonstration power plant with demountable magnets'. *Fusion Engineering and Design*, **100**, pp. 378–405. DOI: `10.1016/j.fusengdes.2015.07.008`.

Summerscales, O. (2023). *Hitting the Jackpot: The Birth of the Monte Carlo Method — LANL*. Owen Summerscales. URL: `https://www.lanl.gov/media/publications/actinide-research-quarterly/1123-hitting-the-jackpot-the-birth-of-the-monte-carlo-method` (visited on 19/10/2025).

Tanabe, T. (2017). *Tritium: Fuel of Fusion Reactors*. Tokyo: Springer.

Tas, H. et al. (1988). 'Liquid breeder materials'. *Journal of Nuclear Materials*, **155-157**, pp. 178–187. DOI: `10.1016/0022-3115(88)90239-5`.

Tosti, S. and Ghirelli, N. (2013). *Tritium in fusion: Production, uses and environmental impact*. Hauppage: Nova Science Publishers.

Valentine, A. et al. (2022). 'Benchmarking of emergent radiation transport codes for fusion neutronics applications'. *Fusion Engineering and Design*, **180**, p. 113197. DOI: `10.1016/j.fusengdes.2022.113197`.

Wandelt, K., ed. (2018). *Encyclopedia of interfacial chemistry: surface science and electrochemistry*. Amsterdam, Netherlands Oxford, UK Cambidge, USA: Elsevier. 1 p.

Wang, X. et al. (2019). 'Current design and R&D progress of the Chinese helium cooled ceramic breeder test blanket system'. *Nucl. Fusion*, **59** (7), pp. 1–8. DOI: `10.1088/1741-4326/ab0c32`.

Wu, Y. (2011). 'Overview of liquid lithium lead breeder blanket program in China'. *Fusion Engineering and Design*, **86** (9), pp. 2343–2346. DOI: `10.1016/j.fusengdes.2010.12.046`.

Zheng, S. and Todd, T. N. (2015). 'Study of impacts on tritium breeding ratio of a fusion DEMO reactor'. *Fusion Engineering and Design*, **98-99**, pp. 1915–1918. DOI: `10.1016/j.fusengdes.2015.06.171`.